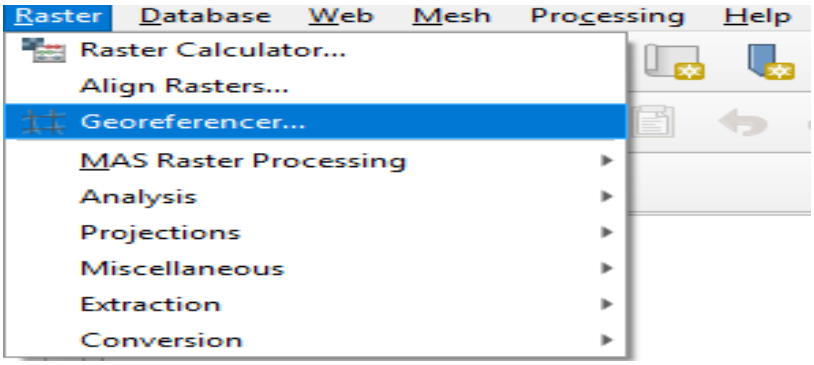
Practical No – 7

Georeferencing Topo Sheets and Scanned Maps Georeferencing Aerial Imagery Digitizing Map Data

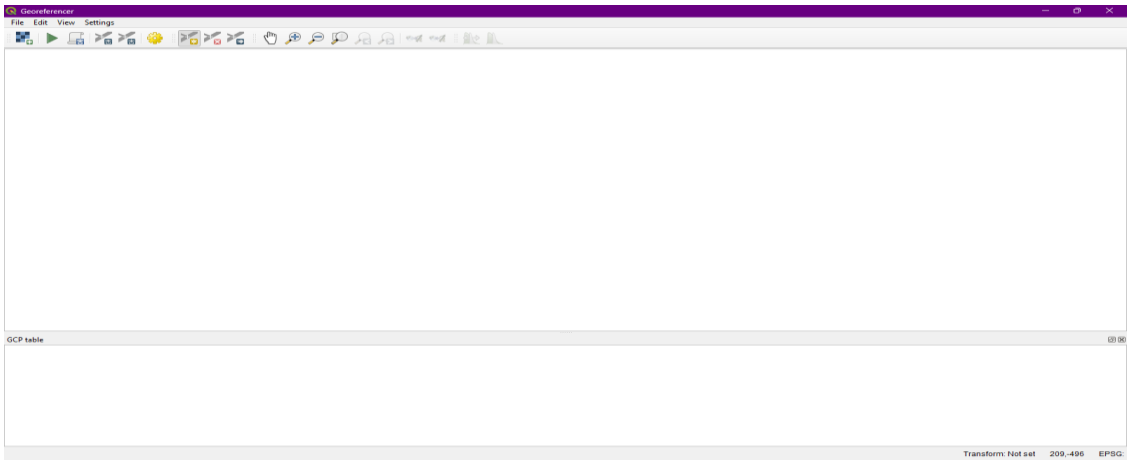
A. Georeferencing Topo Sheets and Scanned Maps

- Go to Layer→Add Layer→Add vector Layer
- Select GIS_Workshop\Manual\Prac06\IND_adm0.shp
- Zoom in to Mumbai region in the layer.

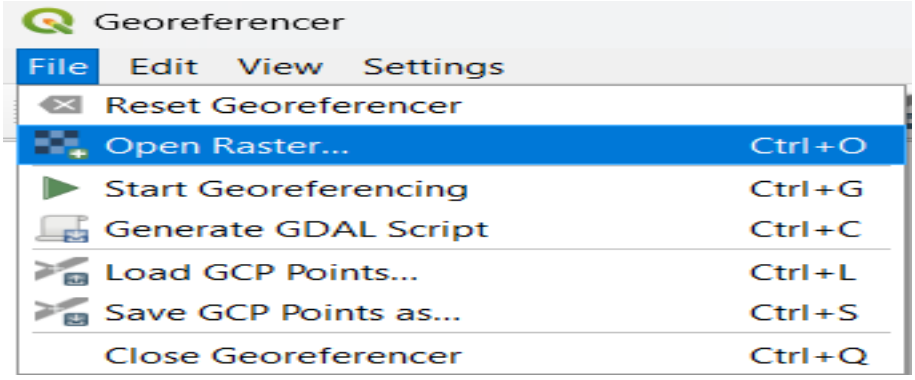
- Go to Plugins→Manage and Install Plugins
- Ensure that #image is checked, if not install georeferencer GDAL plugin.
- Go to Raster→Georeferencer



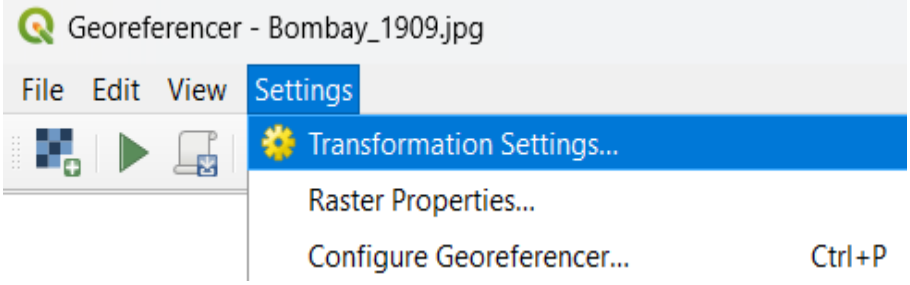
- A new Georeferencer window will open



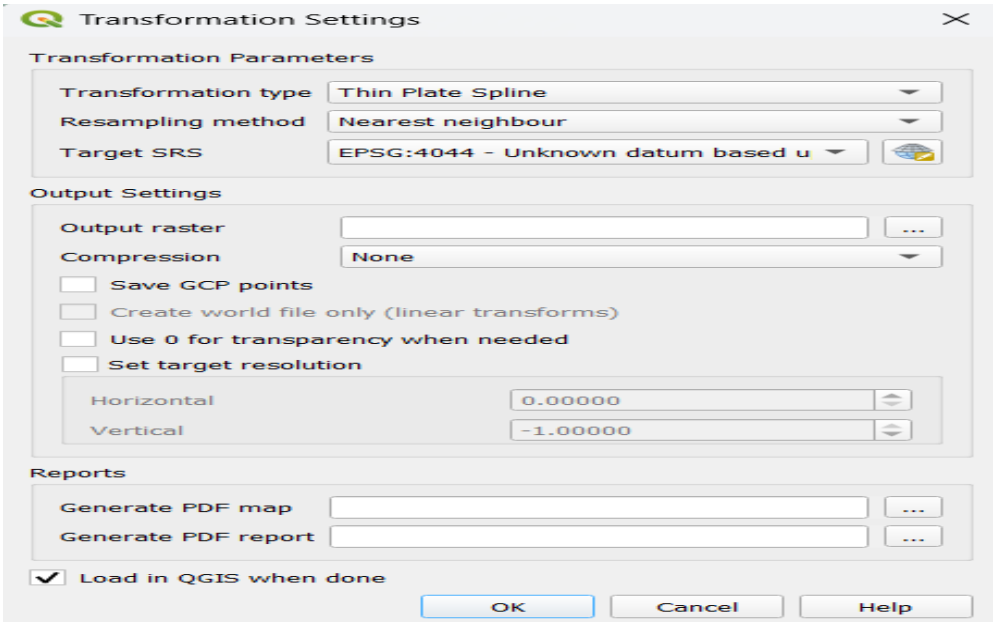
- File→Open Raster



- Select file "1870_southern-india_3975_3071_600.jpg" from project data folder.
- Image
- Go to settings→Transformation Settings



- In the Transformation Settings Windows



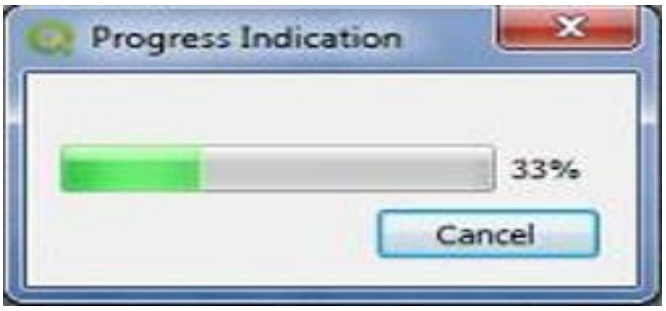
- Select Transformation type→Thin Plate Spline
- Re-sampling Method→Nearest Neighbour
- Target TRS→Everest 1830 datum: EPSG 4044
- Select Output Raster Name and Location
- Check the Load in QGIS When Done Option
- Press "OK
-

- In Georeferencer window Go to Edit→Add Points

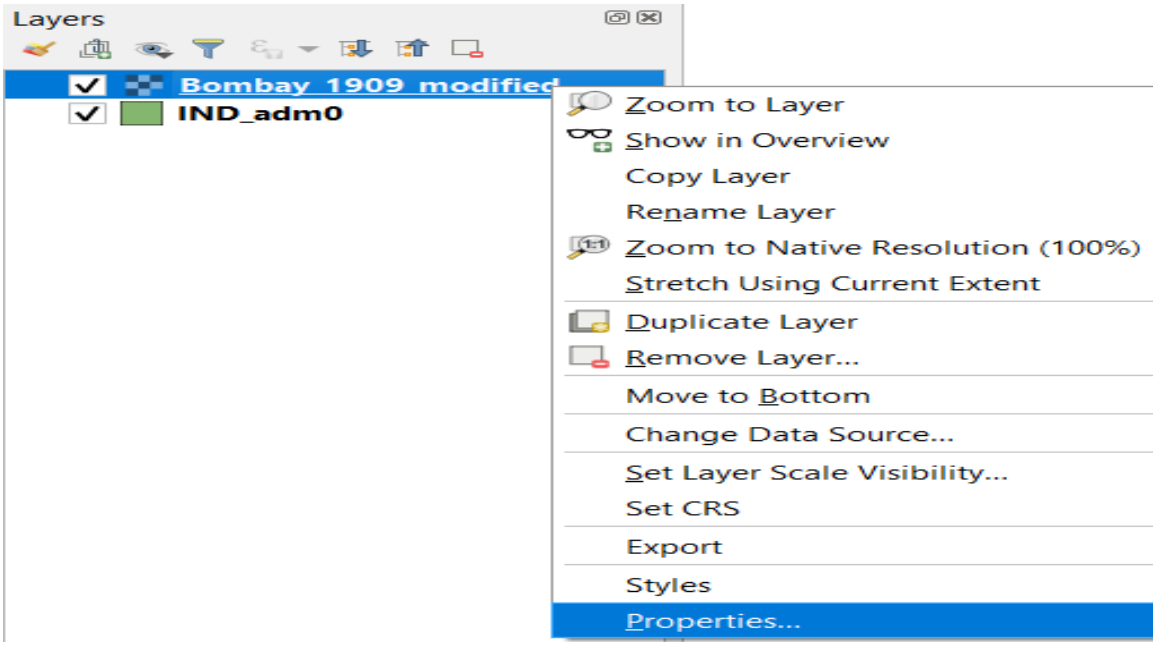
GCP table

Visible	ID	Source X	Source Y	Dest. X	Dest. Y	dX (pixels)	dY (pixels)	Residual (pixels)
☑	0	313.914	-1438.08	72.7915	18.9415	4.93856e-10	5.25915e-10	7.21443e-10
☑	1	925.421	-1482.13	72.8413	18.9178	5.16366e-10	5.32054e-10	7.41428e-10
☑	2	490.111	-880.985	72.8177	19.0436	4.92491e-10	5.23755e-10	7.18935e-10
☑	3	938.377	-844.709	72.8835	19.009	5.30804e-10	5.23301e-10	7.45383e-10

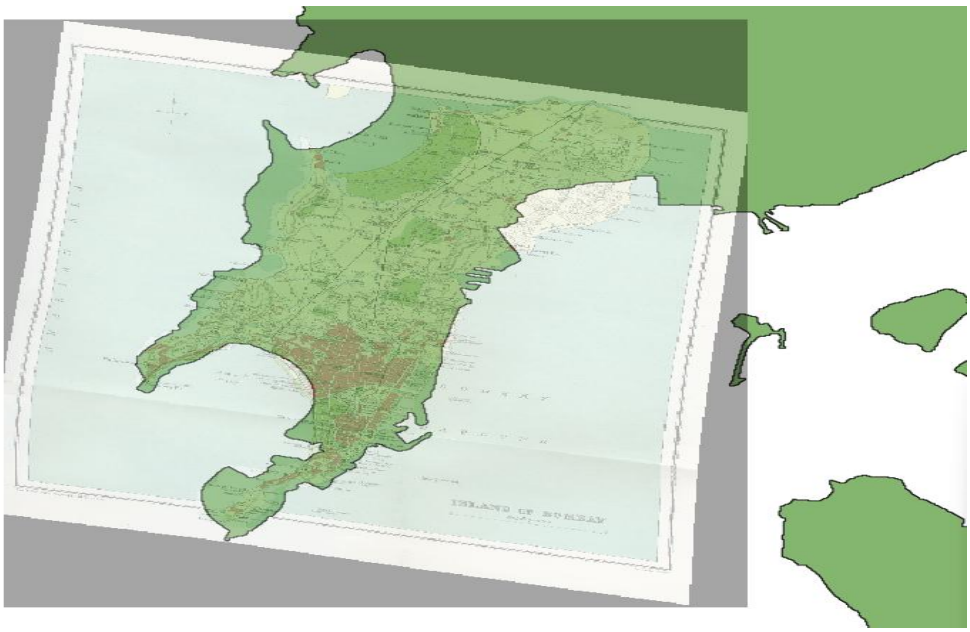
- Select the set of control points.
- Go to, Setting→transformation settings.
- Press “RUN”
- In Georeferencing windows go to→File→Start Georeferencing
- The progress indicator will appear



- The canvas area will now have the scanned map of Mumbai referenced with control points.
- Select the newly added layer in layer in layer panel right click and go to property.\

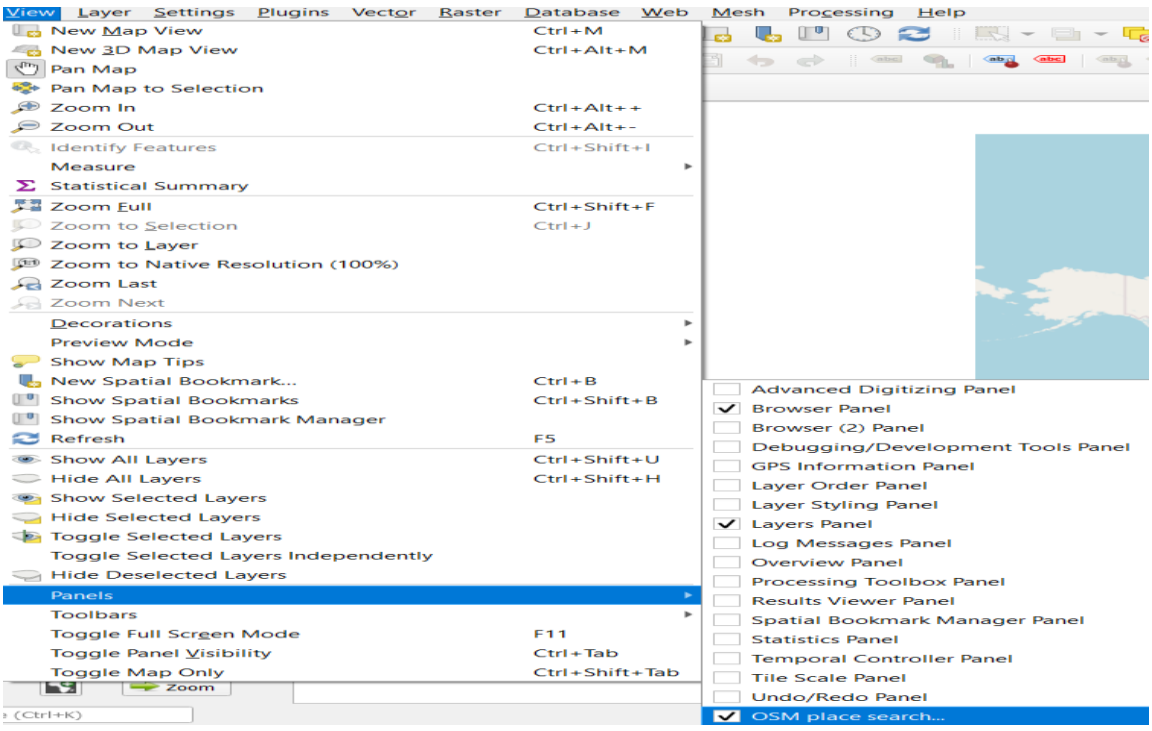


- Set Transparency level of raster layer to appropriate level.

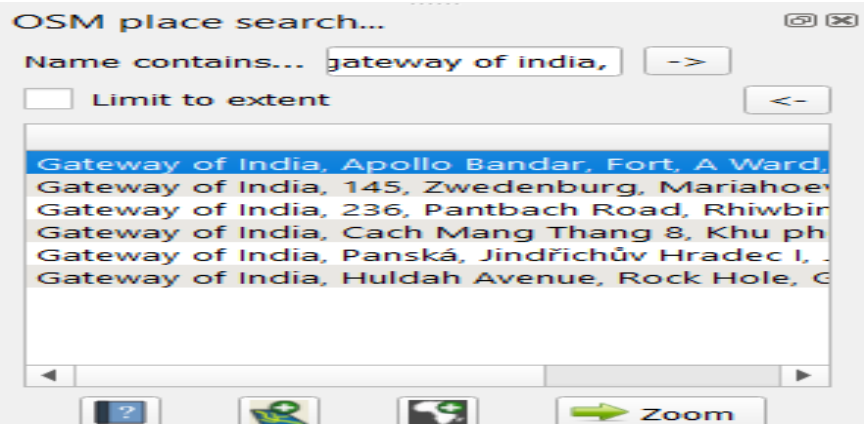


B. Georeferencing Aerial Imagery

- Install plugin OpenStreetMap
- Go to Web Menu→ OpenLayerPlugin→ OpenStreetMap→ OpenStreetMap
- Go to project→Properties→Set CRS to EPSG 3857
- Go to View→Panels→Select OSM Place search

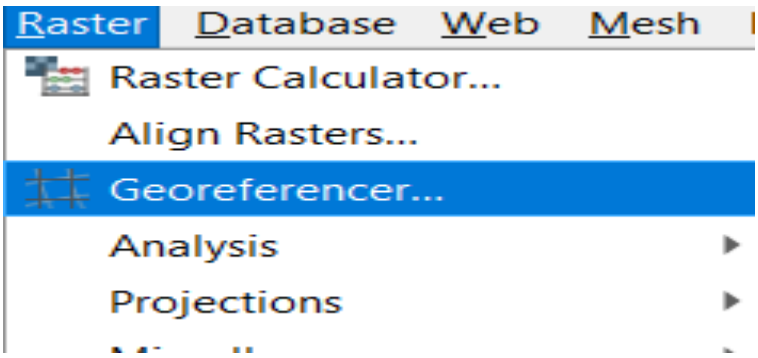


- The Gateway of India, Mumbai is located at 18.92^{0}N 72.83^{0}E
- Search Gateway of India in OSM Search Panel

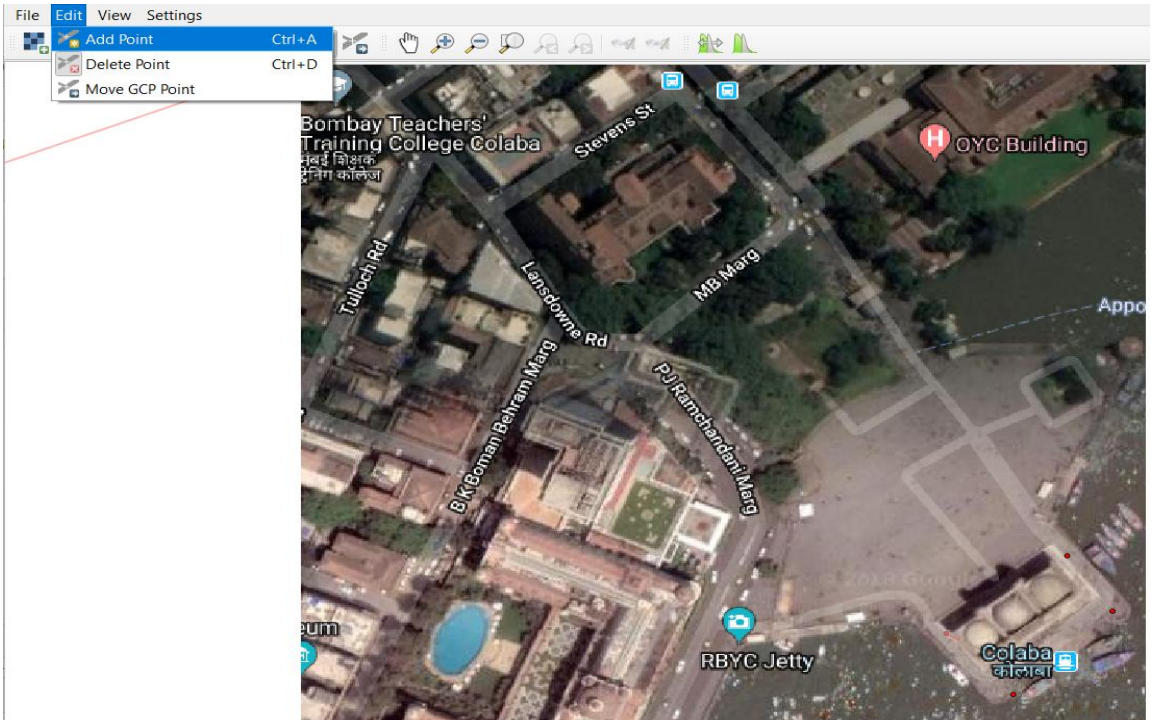


- Zoom in to appropriate level.
- The map will appear like this

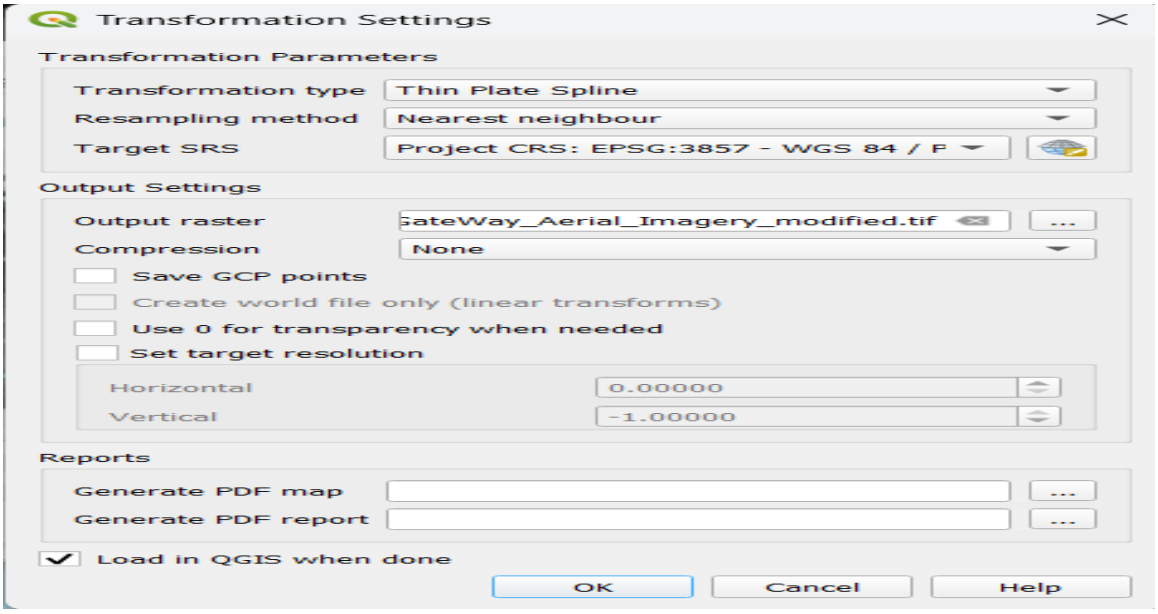
- Go to Raster→Georeferencer



- A new Georeferencer window will open→File→Open Raster→Select file "Gateway_Imagery.tif" from project data folder



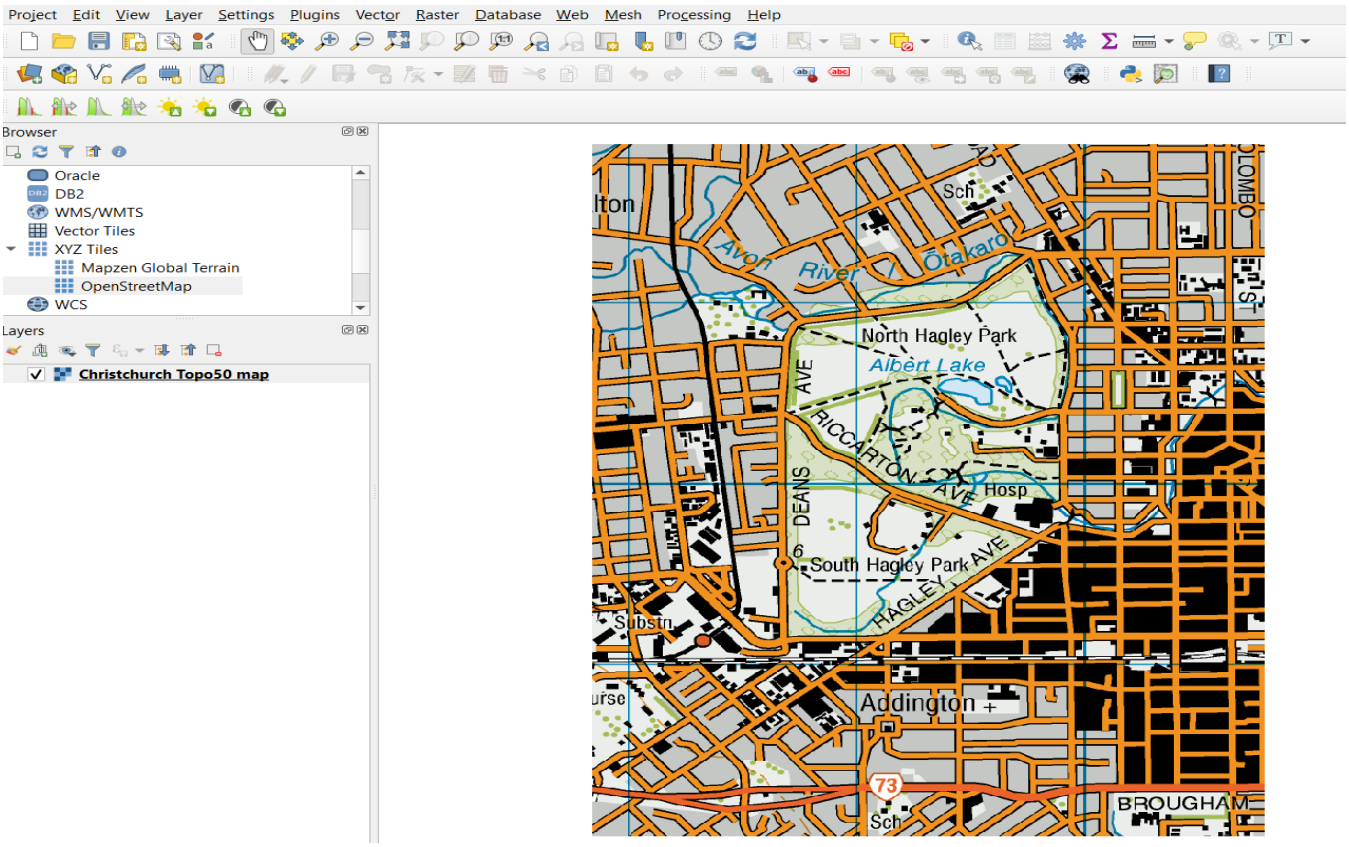
- Go to Edit→Add Point
- Select control points from map (Indicated in red color).
- Go to Setting→Transformation Setting



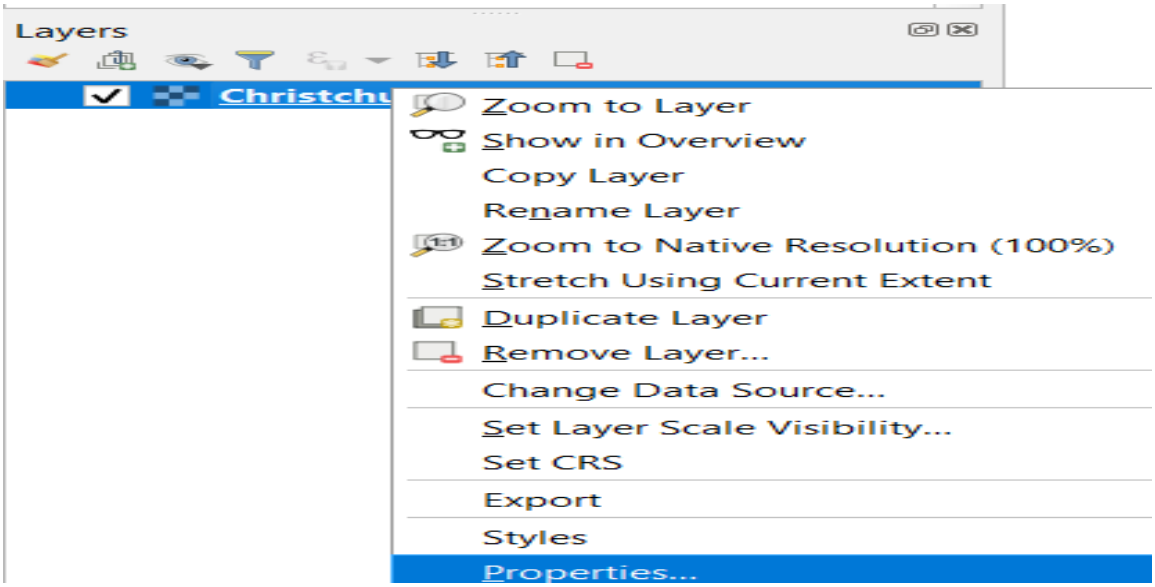
- Go to File→Start Georeferencing or Press the #image button in Georegerencing window.
- The progress indicator will appear

C. Digitizing Map Data

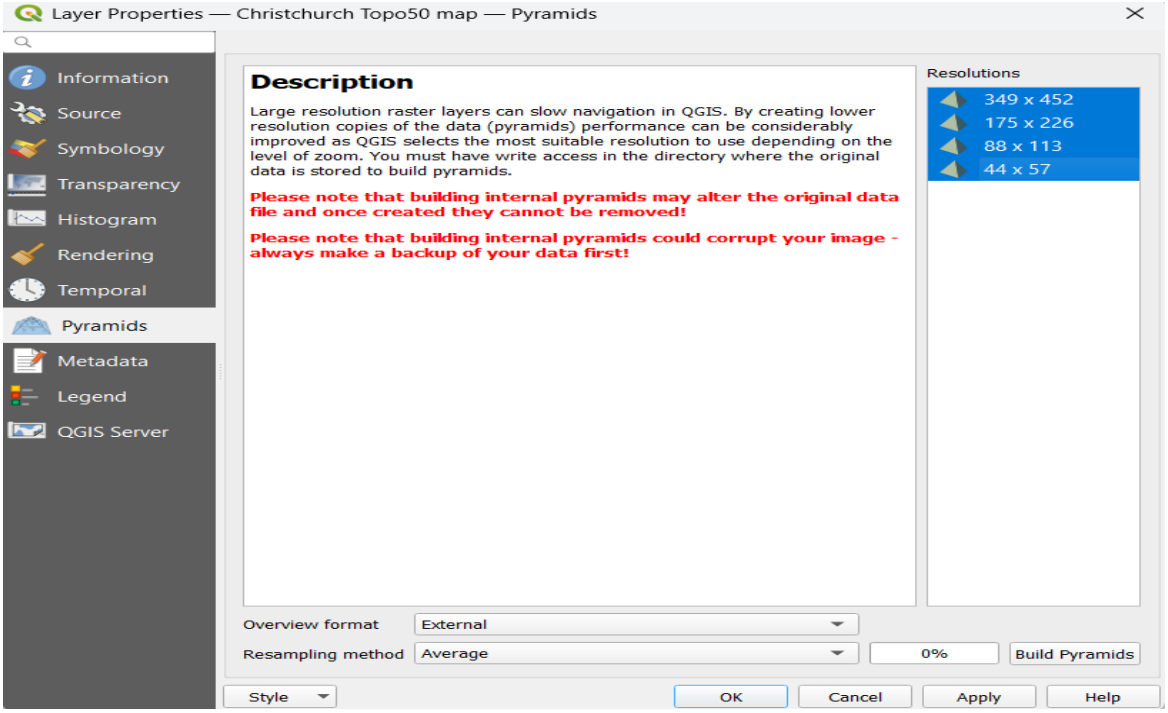
- Go to Layer→Add Raster→Select "Christchurch Topo50map.tif" from project folder.



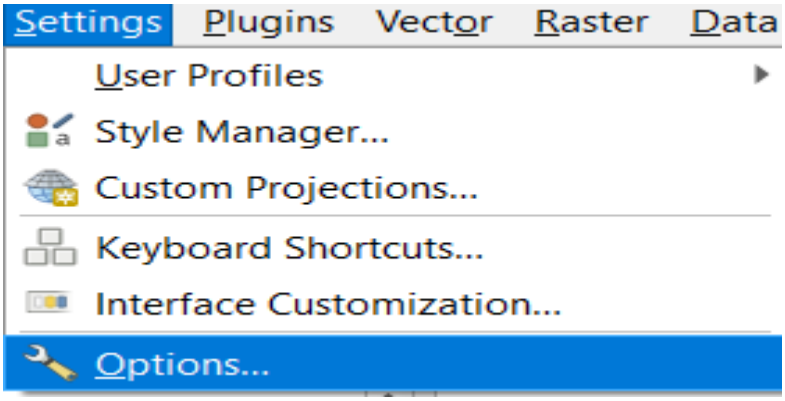
- QGIS offers a simple solution to make raster load much faster by using Image pyrmids.
- Right-click the Christchurch Topo50 map.if layer and select Properties.



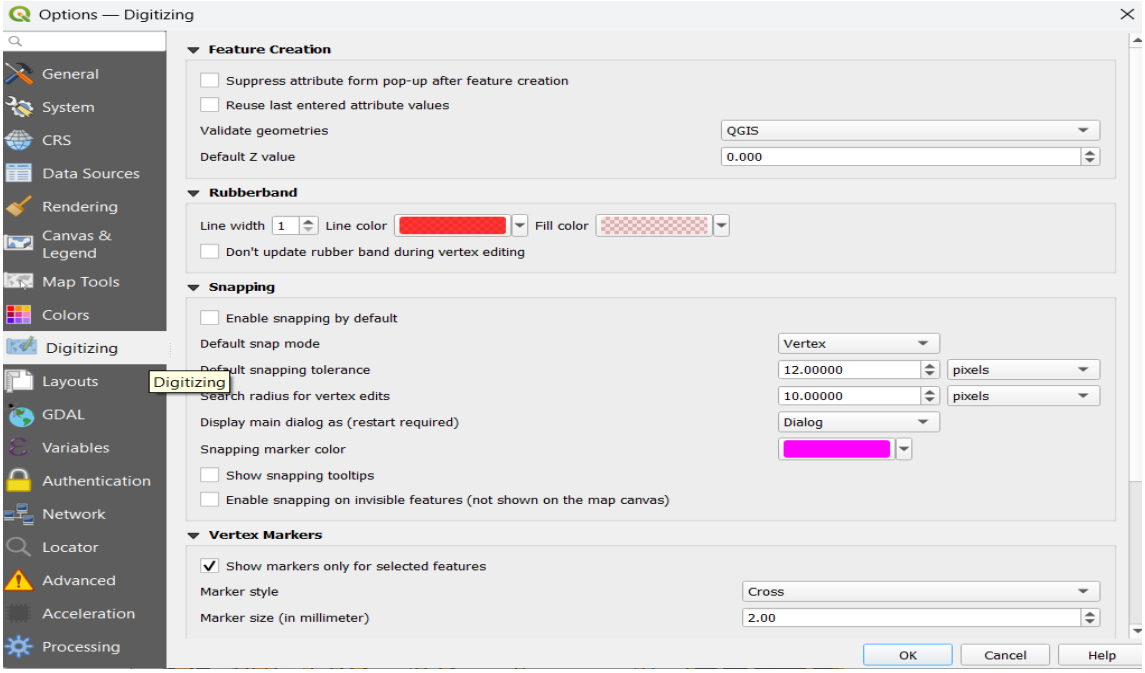
- Select Properties. Choose the Pyramids tab. Hold the Ctrl key and select all the resolutions offered in the resolutions panel.



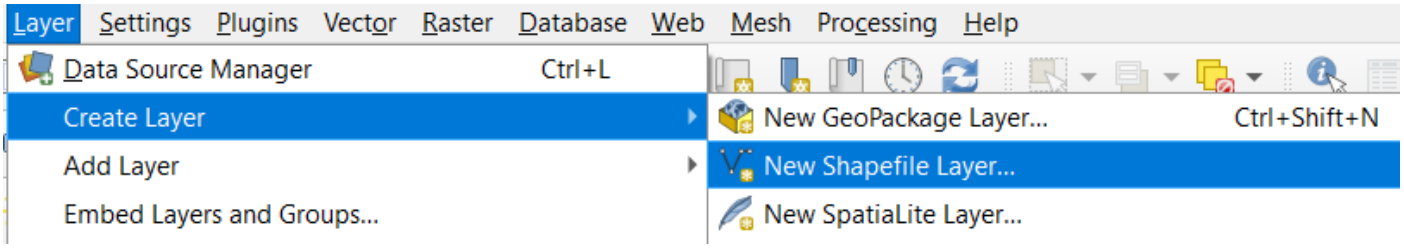
- Click Build pyramids. Then Click OK
- Go to Settings→Options….Select the Digitizing tab in the Options dialog.



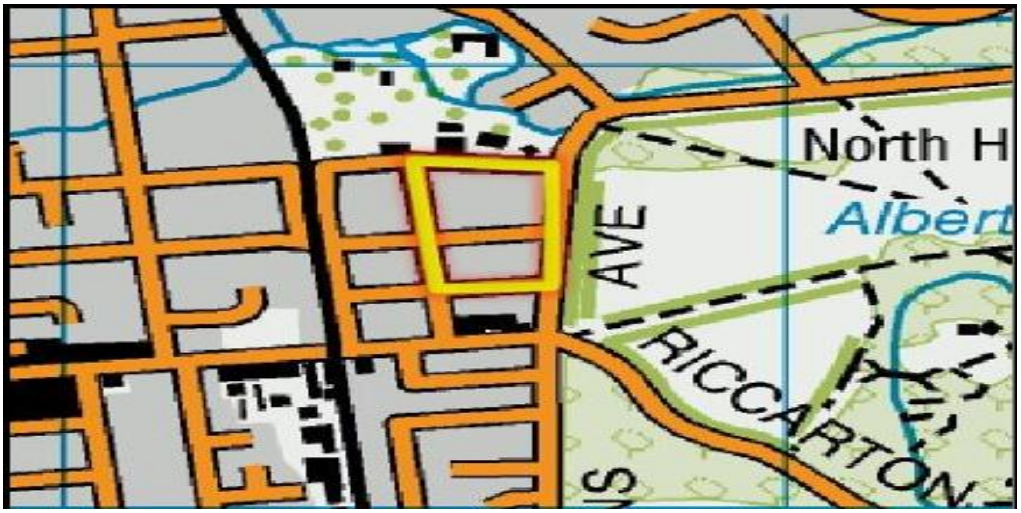
- Set the Default snap mode to vertex and segment.



- Press OK.
- Go to Layer→Add Layer→Add Spatialite Layer.



- Select the name and location for spatial database eg. "GIS_Workshop\Practicals\Practical_06\C\MySpatialDataBase.sqlite".
  - Name the Layer as "Digitized_Road
  - Set Geometry type as "Line"
  - Set CRS EPSG:4167—NZGD2000
- Image
- Add "Name" fields using "Add to Fields List".
- On Layer Panel Right Click on Digitze_Road, Select the Style tab in the Layer Properties dialog.



- After creating a new Spatiality layer→Select Digitized_Garden layer in layer Panel and click on Toggle Editing button and then Add Polygon Feature button on Tool bar.
- After two gardens to the region by adding polygon.

- Styling and labelling option in layer